Khachiyan's Ellipsoid Algorithm

for Linear Programming

CSE 461 Algorithm Engineering

Bangladesh University of Engineering and Technology (BUET)

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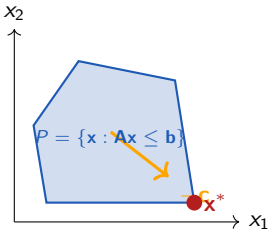
Linear Programming: The Problem

Standard Form LP

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^n} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

where $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^n$.



Feasible polytope P and optimal vertex $\mathbf{x}^*$

Feasibility Version

Does there exist $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} \leq \mathbf{b}$?

WLOG: Optimization $\Leftrightarrow$ Feasibility

Key parameters:

- n = number of variables
- m = number of constraints

Before 1979: The Simplex Dilemma

Simplex Method (Dantzig, 1947)

- Walks along *vertices* of the feasibility polytope
- **Extremely fast in practice**
- **Worst-case exponential time**
- Klee-Minty (1972): 2^n pivots on a specially crafted instance

Klee-Minty: Simplex visits 2^n vertices

The Open Question (1972–1978)

"Is Linear Programming in $\mathcal{P}$?"

Unknown for 30+ years after LP was formulated. Crucial for theory: LP is complete for P under log-space reductions.

Timeline

1947: Simplex (Dantzig)
1972: Klee-Minty example
1979: Ellipsoid (Khachiyan)
1984: Interior-Point (Karmarkar)

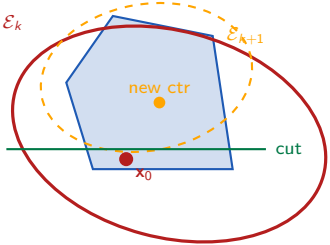
Motivation: Why the Ellipsoid Method?

Leonid Khachiyan (1979) adapted the ellipsoid method (Yudin & Nemirovski, 1976) from convex optimisation to LP and proved:

Historic Result

$$\text{LP} \in \mathcal{P} \quad \text{(polynomial time solvable)}$$

Running time: $\mathcal{O}(n^6 L^2)$ bit operations

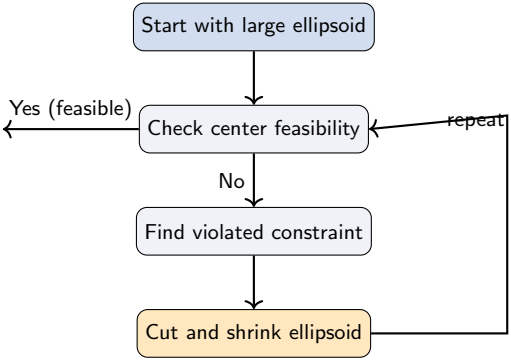


Ellipsoid shrinks around polytope P

Core idea:

- Enclose the feasible region in an *ellipsoid* $\mathcal{E}$
- If centre $\mathbf{x}_0 \notin P$: find a violated constraint
- Replace $\mathcal{E}$ with a *smaller* ellipsoid covering half of $\mathcal{E}$ containing P
- Repeat until feasibility confirmed or $\mathcal{E}$ is too small

Big Picture: How the Algorithm Works



Core Idea

Each iteration removes a **half-space** and replaces it with a **smaller ellipsoid**, ensuring convergence.

Notation & Definitions

Ellipsoid

An *ellipsoid* in $\mathbb{R}^n$ centred at $\mathbf{z}$ with positive-definite matrix $B \succ 0$:

$$\mathcal{E}(\mathbf{z}, B) = \{\mathbf{x} \in \mathbb{R}^n \mid (\mathbf{x} - \mathbf{z})^\top B^{-1}(\mathbf{x} - \mathbf{z}) \leq 1\}$$

- **Centre**: $\mathbf{z} \in \mathbb{R}^n$
- **Shape matrix**: $B \in \mathbb{R}^{n \times n}$, $B \succ 0$
- **Volume**: $\mathrm{vol}(\mathcal{E}) = \det(B)^{1/2} \cdot \omega_n$
- ω_n = volume of unit ball in $\mathbb{R}^n$

Feasible Polytope

$$P = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{A}\mathbf{x} \leq \mathbf{b}\}, \quad \mathbf{A} \in \mathbb{R}^{m \times n}$$

Key Symbols

Symbol	Meaning
n	Dimension (no. of variables)
m	No. of constraints
L	Bit-length of input
$\mathbf{z}_k$	Centre of k-th ellipsoid
B_k	Shape matrix of k-th ellipsoid
$\mathbf{a}_i$	i-th row of $\mathbf{A}$ (constraint normal)
$\tilde{\mathbf{a}}$	Normalised constraint: $\mathbf{a}/\lVert\mathbf{a}\rVert$
δ	Violation threshold ($\approx 2^{-L}$)
N	Max iterations
R	Initial ball radius
r	Inner ball radius

Algorithm: Pseudocode

Input : $\mathbf{A} \in \mathbb{Q}^{m \times n}$, $\mathbf{b} \in \mathbb{Q}^m$, L (bit-length)
Output: $\mathbf{x}^* \in P$ or INFEASIBLE

$R \leftarrow 2^L$; `// initial ball radius`
$\mathbf{z}_0 \leftarrow \mathbf{0}$;
$B_0 \leftarrow R^2 \cdot I_n$; `// initial sphere`
$N \leftarrow 2n(n+1)L$; `// iteration budget`

for $k = 0, 1, 2, \ldots, N$ **do**
  if $\mathbf{z}_k \in P$ (all constraints satisfied) **then**
    return $\mathbf{z}_k$; `// feasible point found`
  else
    Find violated constraint i: $\mathbf{a}_i^\top \mathbf{z}_k > b_i$;
    $\tilde{\mathbf{a}} \leftarrow \mathbf{a}_i \,/\, \|\mathbf{a}_i\|$;
    `// Update centre`
    $\mathbf{z}_{k+1} \leftarrow \mathbf{z}_k - \dfrac{1}{n+1} B_k \tilde{\mathbf{a}}$;
    `// Update shape matrix`
    $B_{k+1} \leftarrow \dfrac{n^2}{n^2-1}\left(B_k - \dfrac{2}{n+1} B_k \tilde{\mathbf{a}}\tilde{\mathbf{a}}^\top B_k\right)$;
  end
end

Intuition for each step:

1. **Initialise** with a ball large enough to contain any rational solution of P (if one exists): radius $R = 2^L$
2. **Check** if the current centre is feasible
3. **Cut**: If not, the violated half-space separates $\mathbf{z}_k$ from all of P — half of $\mathcal{E}_k$ can be discarded
4. **Fit**: Compute the *minimum-volume ellipsoid* over the remaining half-ellipsoid $\Rightarrow \mathcal{E}_{k+1}$
5. **Terminate**: Either find a feasible point or declare infeasibility when the ellipsoid shrinks below 2^{-nL}

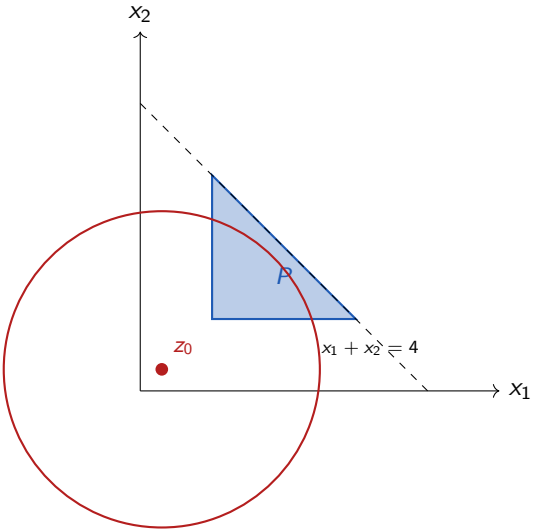
Illustrative Example: Geometry

Feasible Region

$$x_1 + x_2 \leq 4$$
$$x_1 \geq 1$$
$$x_2 \geq 1$$

Interpretation

- Blue region = feasible set P
- Red circle = initial ellipsoid
- Center starts at $(0, 0)$ (infeasible)



Illustrative Example: Iterations

Iteration 0

- Start at $(0, 0)$
- Violates constraints:

$$x_1 \geq 1, \quad x_2 \geq 1$$

Iteration 1

- Apply cutting plane
- Update ellipsoid
- Center shifts toward feasible region

What is happening?

- Infeasible half is removed
- Search region shrinks
- Center moves closer to P

Final Outcome

After several iterations:

$$z_k \in P \Rightarrow \text{feasible solution found}$$

Ellipsoid Update Rule — Derivation

Setup

Given $\mathcal{E}_k = \mathcal{E}(\mathbf{z}_k, B_k)$ and violated half-space

$$H^- = \{\mathbf{x} : \tilde{\mathbf{a}}^\top \mathbf{x} \leq \tilde{\mathbf{a}}^\top \mathbf{z}_k\}$$

Update Formulas

$$\mathbf{z}_{k+1} = \mathbf{z}_k - \frac{1}{n+1} B_k \tilde{\mathbf{a}}$$

$$B_{k+1} = \frac{n^2}{n^2-1}\left(B_k - \frac{2}{n+1} B_k \tilde{\mathbf{a}}\tilde{\mathbf{a}}^\top B_k\right)$$

Notes:

- $\tilde{\mathbf{a}} = \mathbf{a}/\|\mathbf{a}\|$
- Rank-1 update
- Cost: $\mathcal{O}(n^2)$

Volume Reduction

$$\frac{\mathrm{vol}(\mathcal{E}_{k+1})}{\mathrm{vol}(\mathcal{E}_k)} = \left(\frac{n^2}{n^2-1}\right)^{n/2} \cdot \frac{1}{n+1} \leq e^{-1/(2n)}$$

Proof Sketch

- Matrix determinant lemma

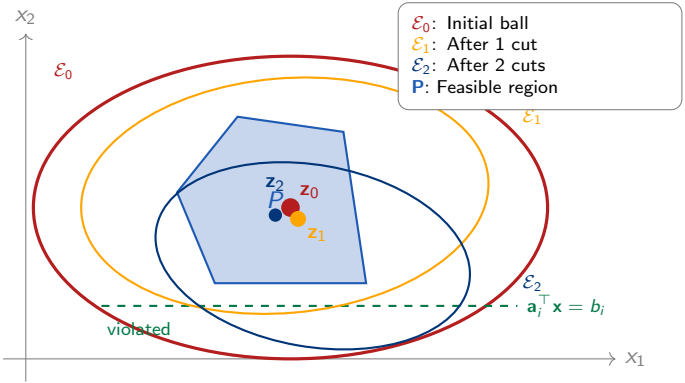
$$\det(I - \alpha u u^\top) = 1 - \alpha \|u\|^2$$

- $\mathrm{vol}(\mathcal{E}) \propto \det(B)^{1/2}$
- exponential decay

Key Result

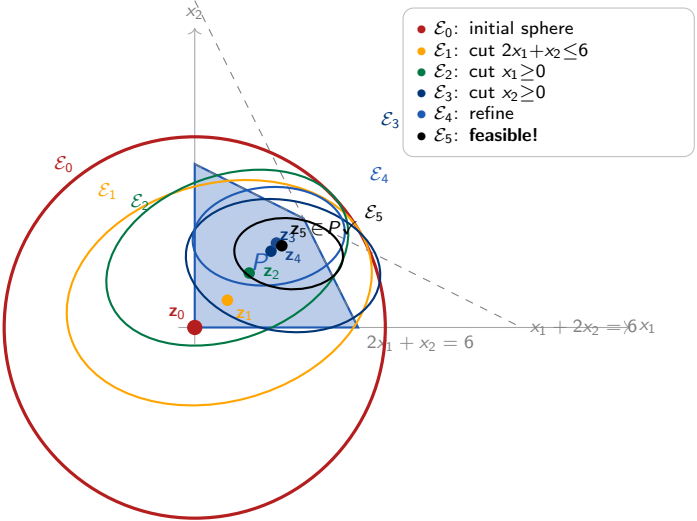
Volume shrinks by factor $e^{-1/(2n)}$ per iteration

Geometric Intuition — Step by Step



Simulation: 5 Iterations in 2D

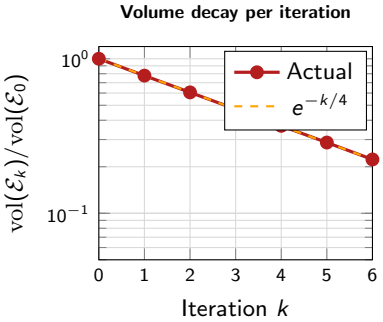
LP: $x_1 + 2x_2 \leq 6,\ 2x_1 + x_2 \leq 6,\ x_1, x_2 \geq 0.$ Initial ellipsoid: sphere of radius 8.



Numerical Trace of Iterations

Iteration Table ($n = 2$, toy instance)

k	$z_{k,1}$	$z_{k,2}$	$\mathrm{vol}(\mathcal{E}_k)/\mathrm{vol}(\mathcal{E}_0)$	Status
0	0.000	0.000	1.0000	infeasible
1	0.600	0.400	$e^{-1/4}$	infeasible
2	0.980	0.810	$e^{-1/2}$	infeasible
3	1.240	1.120	$e^{-3/4}$	infeasible
4	1.450	1.380	e^{-1}	infeasible
5	1.610	1.490	$e^{-5/4}$	**feasible**



Log-linear decay

$$\ln \mathrm{vol}(\mathcal{E}_k) \approx \ln \mathrm{vol}(\mathcal{E}_0) - \frac{k}{2n}$$

Volume decays as $e^{-k/(2n)}$ per step ($n = 2 \Rightarrow e^{-k/4}$).

For general n: need $N = 2n^2 L$ steps before declaring infeasibility.

Volume Reduction Lemma

Theorem (Volume Reduction)

Let $\mathcal{E}_k = \mathcal{E}(\mathbf{z}_k, B_k)$. *Then*

$$\frac{\mathrm{vol}(\mathcal{E}_{k+1})}{\mathrm{vol}(\mathcal{E}_k)} = \left(\frac{n^2}{n^2-1}\right)^{n/2} \cdot \frac{1}{n+1} \leq e^{-1/(2n)}$$

Interpretation

- Each step removes **half-space**
- New ellipsoid tightly fits remaining region
- Volume decreases **exponentially**

Key Insight

After k iterations:

$$\mathrm{vol}(\mathcal{E}_k) \leq \mathrm{vol}(\mathcal{E}_0) \cdot e^{-k/(2n)}$$

Volume Reduction: Proof & Termination

Proof Sketch

1. Apply matrix determinant lemma:

$$\det(I - \alpha uu^\top) = 1 - \alpha\|u\|^2$$

2. Volume relation:

$$\mathrm{vol}(\mathcal{E}) \propto \det(B)^{1/2}$$

3. Combine $\Rightarrow$ exponential decay

Termination Guarantee

After $N = 2n(n+1)L$ iterations:

$$\mathrm{vol}(\mathcal{E}_N) < 2^{-nL}$$

Conclusion:

- If $P \neq \emptyset \rightarrow$ feasible point found
- Otherwise $\rightarrow$ declare infeasible

Complexity Analysis

Per-Iteration Cost

$$\mathcal{O}(mn + n^2)$$

- Feasibility check: $\mathcal{O}(mn)$
- Matrix update: $\mathcal{O}(n^2)$

Iterations

$$N = \mathcal{O}(n^2 L)$$

Arithmetic Complexity

$$T_{\text{arith}} = \mathcal{O}(n^4 L)$$

Bit Complexity

$$T_{\text{bit}} = \mathcal{O}(n^6 L^2)$$

Key Result

$$\textbf{LP} \in \mathcal{P}$$

Algorithm Comparison

Property	Simplex	Ellipsoid	Interior-Point
Year	1947	1979	1984
Author	Dantzig	Khachiyan	Karmarkar
Worst-case time	$\exp(n)$	$\mathcal{O}(n^6 L^2)$	$\mathcal{O}(n^{3.5} L^2)$
Practical speed	**Very fast**	Slow	**Fast**
In $\mathcal{P}$?	No	**Yes (first!)**	Yes
Path type	Vertex walk	Volumetric shrink	Interior trajectory
Works for CVX?	No	Yes	Yes
Used in practice?	**Yes**	Rarely	**Yes**

Key Insight: The Ellipsoid Method is a *theoretical* breakthrough proving LP $\in \mathcal{P}$, but interior-point methods dominate practice. Its real legacy is the *subgradient framework* used in combinatorial optimisation.

Legacy, Extensions & Open Problems

Legacy of the Ellipsoid Method

- First **polynomial-time** LP algorithm
- Works for any **convex** feasibility problem with a *separation oracle* — even exponentially many constraints
- Powers polynomial algorithms for:
  - Maximum matching in general graphs
  - Submodular function minimisation
  - Semi-definite programming
  - Network flow with complex constraints
- Inspired Karmarkar's interior-point method (1984)

Open Problems

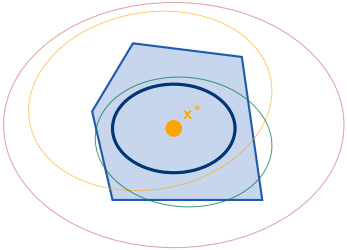
1. **Strongly polynomial LP?**
   Running time independent of L?
   Smale's 9th problem (millennium list)
2. **Average-case Simplex?**
   Why is Simplex fast in practice?
3. **LP lower bounds?**
   Is $\Omega(n^2)$ iterations necessary?
4. **Better SDP algorithms?**
   Current best: $\mathcal{O}(n^{4.5}L)$

Separation Oracle

Given $\mathbf{x}$, either confirm $\mathbf{x} \in P$ or return a hyperplane separating $\mathbf{x}$ from P.
Ellipsoid runs in $\mathrm{poly}(n, m, L)$ oracle calls.

Summary

1. **LP** is the fundamental problem of optimising a linear objective over a polytope, parameterised by n, m, L.
2. **Before 1979**, Simplex was exponential in the worst case; LP's membership in $\mathcal{P}$ was open.
3. **Khachiyan (1979)** showed LP $\in \mathcal{P}$ using the Ellipsoid Method with complexity $\mathcal{O}(n^6 L^2)$.
4. Each iteration replaces the enclosing ellipsoid with a **smaller** one using the closed-form update $(\mathbf{z}_{k+1}, B_{k+1})$.
5. Volume decreases by factor $e^{-1/(2n)}$ per step, guaranteeing termination in $\mathcal{O}(n^2 L)$ iterations.
6. Its greatest legacy is the **separation oracle** paradigm enabling poly-time algorithms for a vast class of combinatorial problems.



Ellipsoids converge to P

LP $\in \mathcal{P}$ — Khachiyan, 1979

Complexity: $\mathcal{O}(n^6 L^2)$

Thank You!

Questions?